AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 15 · STUDENT LESSON

Choose Informative Questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

List possibilities. Count both answers. Recount after a change.

Choose a question under a declared worst-case criterion and explain when a previously useful clue becomes constant.

Keep one habit: Eight-card game → largest branch comparison → restricted set → new independent comparison.

MAIN LESSON · SECTION 1

Which question leaves the fewest possibilities in the worst case?

> Declare what useful means for this guessing game.

This week's goals

I can compare questions by their largest remaining group.

I can recount the split after the possibility set changes.

I can state assumptions and limits of a question-choice rule.

Retrieve a useful question from last week

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Your end goal

By the end, I can choose a question that separates possibilities rather than repeating a constant clue.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

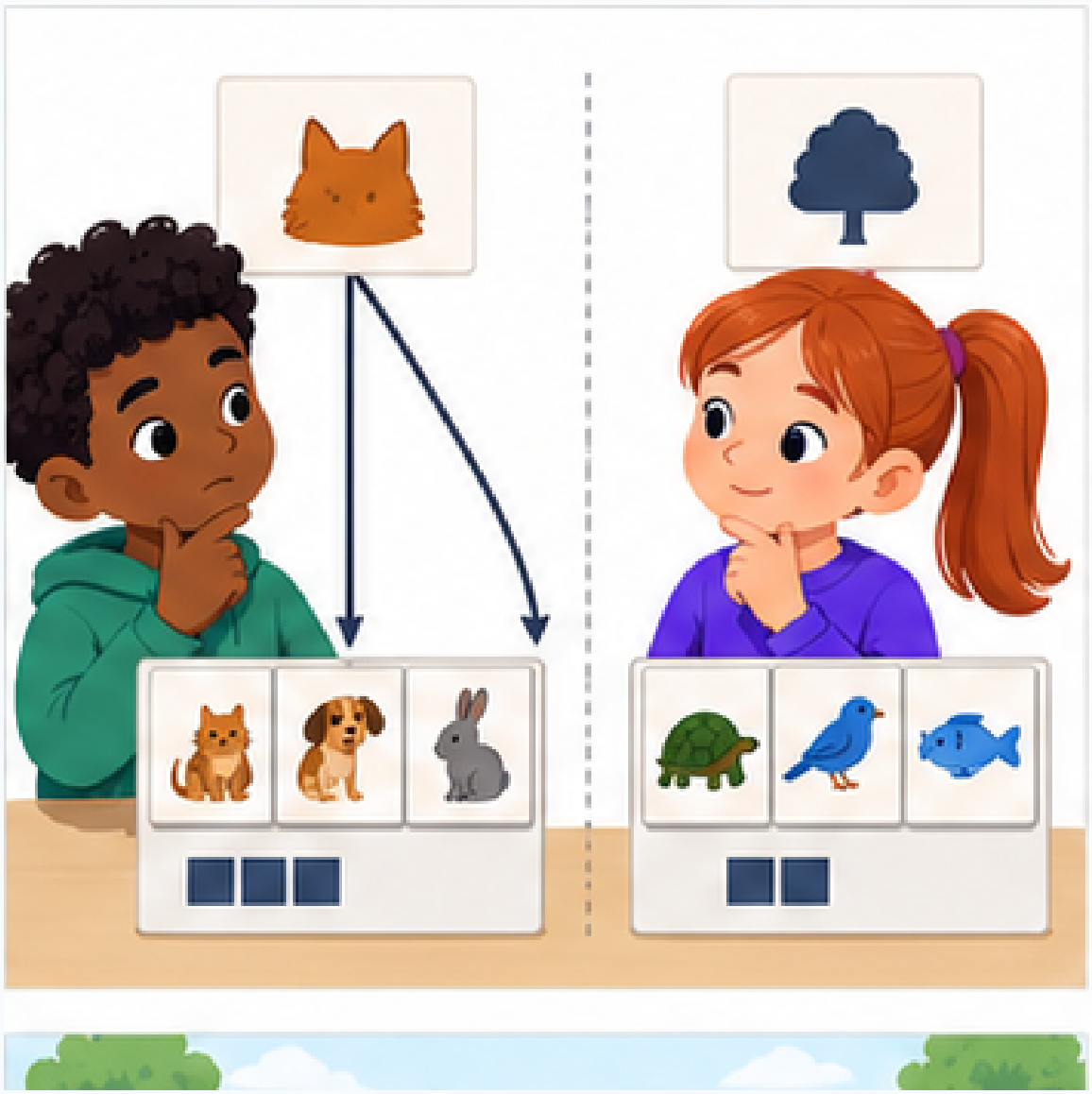
Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 15: Choose **Informative** Questions

A strong question eliminates many possibilities without relying on accidents.
</details>



THINKER MISSION

Compare two first questions by **remaining possibilities**.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Why does a cutoff with zero **training** errors still need validation on separate examples?

This week’s end goal

MAIN LESSON · SECTION 2

Read the complete eight-card possibility set

> All fields and the decision criterion are supplied.

The hidden-target game and worst-case rule

Imagine one hidden target among eight fictional catalogue cards. All eight are still possible. A truthful yes/no answer lets you keep only cards matching that answer. Compare questions by their worst-case remaining group: the larger of the Yes and No counts. Choose the question with the smallest such number. This is a stated identification-game criterion, not a training-error or probability calculation.

All eight fictional cards

Card	Subject	Binding	Language
M1	mystery	paperback	English
M2	mystery	paperback	English
M3	mystery	paperback	English
M4	mystery	hardback	English
S1	science	paperback	English
S2	science	paperback	English
S3	science	paperback	English

Card	Subject	Binding	Language
S4	science	hardback	English

The questions, completeness check and tie rule

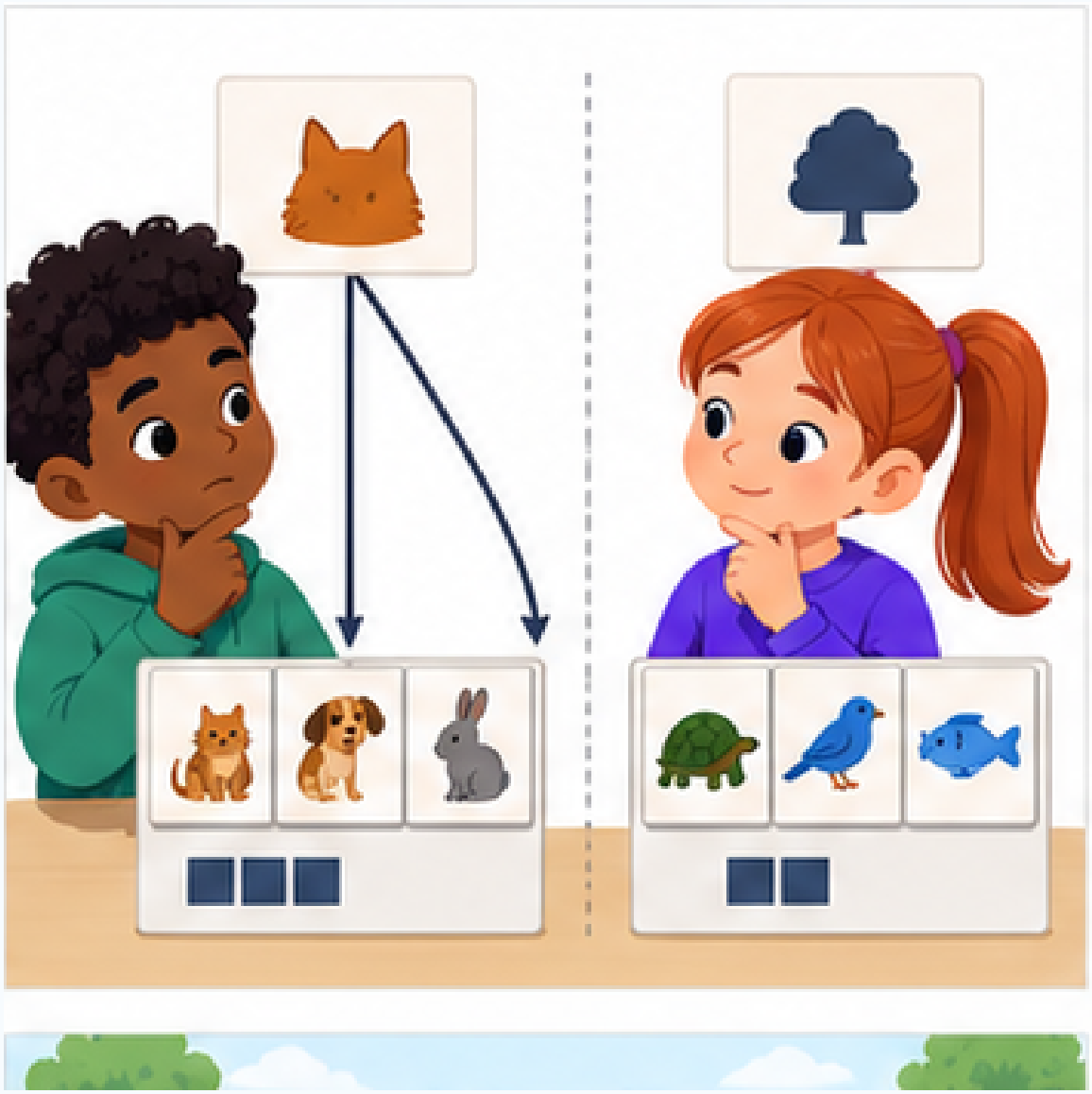
Try mystery?, paperback? and English? in that order. The questions test the printed subject, binding and language fields. Every card enters one answer group; Yes+No must equal the current number of possibilities. A constant clue makes one group empty and leaves every possible target in the other. If the worst-case counts tie, keep the first question for this exercise. One useful question need not identify a unique card.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use choose **informative** questions to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A decision tree asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump. This retained Week 14 review fits a question and its **leaf prediction**s from training labels. The main Week 15 guessing-game task instead compares the largest group of remaining possible target cards.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear

Row	Near	Tall	Label
5	0	0	clear
6	0	1	blocked

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

MAIN LESSON · SECTION 3

Count both answer branches and compare their larger sizes

> This code audits questions; it does not fit a classifier.

Predict before executing the complete audit

```
cards = [("M1", "mystery", "paperback", "English"),
         ("M2", "mystery", "paperback", "English"),
         ("M3", "mystery", "paperback", "English"),
         ("M4", "mystery", "hardback", "English"),
         ("S1", "science", "paperback", "English"),
         ("S2", "science", "paperback", "English"),
         ("S3", "science", "paperback", "English"),
         ("S4", "science", "hardback", "English")]
questions = [("mystery?", 1, "mystery"),
             ("paperback?", 2, "paperback"),
             ("English?", 3, "English")]
def audit(possibilities):
    best = None
    best_worst = len(possibilities) + 1
    for name, column, value in questions:
        yes = sum(card[column] == value for card in possibilities)
        no = len(possibilities) - yes
        worst = max(yes, no)
```

```
            print(name, yes, no, worst)
            if worst < best_worst:
                best, best_worst = name, worst
        return best
    print("chosen", audit(cards))
    remaining = [card for card in cards if card[1] == "mystery"]
    print("after mystery Yes", audit(remaining))
```

Understand the count and candidate comparison

The tuple fields are ID0, subject1, binding2, language3. Each question stores its name, field position and wanted value. Equality gives True for a Yes card and False for a No card; sum counts True as1 and False as0. no is the remaining count and max chooses the larger branch. The strict < keeps the first candidate when scores tie. The final list comprehension keeps the four mystery cards before recounting. The audit assumes a nonempty possibility set with these complete, trusted fields.

Interpret the original and changed-set results

For all eight cards the question rows print mystery?4 4 4; paperback?6 2 6; English?8 0 8, then chosen mystery?. After a truthful Yes to mystery?, the four remaining cards give mystery?4 0 4; paperback?3 1 3; English?4 0 4, then after mystery Yes paperback?. That second comparison uses the explicitly supplied joint table, not just the original marginal totals.

▼ Week 14 review · Earlier fitted-tree program

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
```

```
                    if row[feature] == value]
            leaves[value] = majority(labels)
        errors = 0
        for row in rows:
            if leaves[row[feature]] != row[2]:
                errors = errors + 1
        print(feature, errors)
        if errors < best_errors:
            best_errors = errors
            tree = (feature, leaves)
    print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat **split** selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

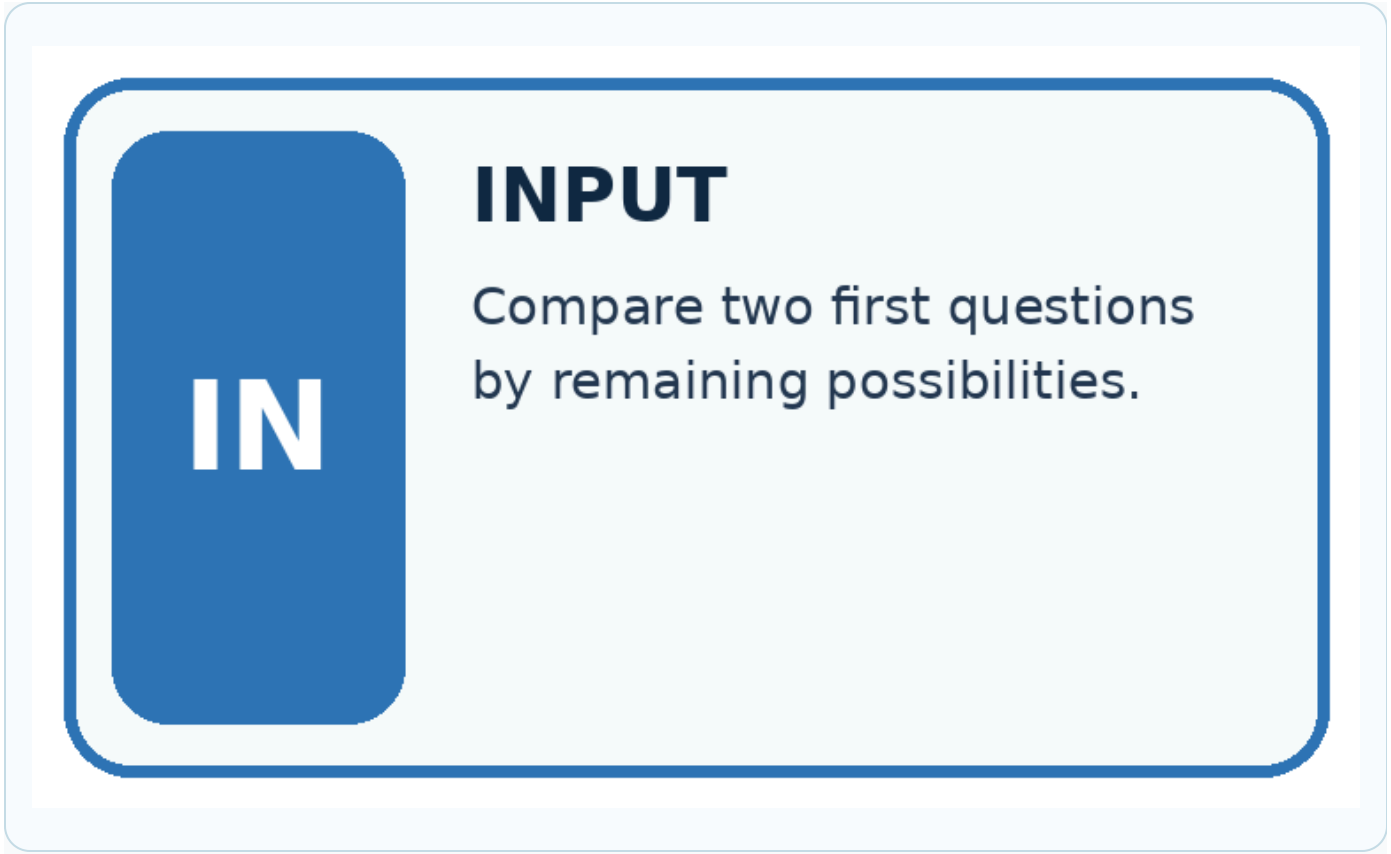
Keeping the scope small lets you inspect every decision.

▼ Earlier wording, alternatives and picture transcripts

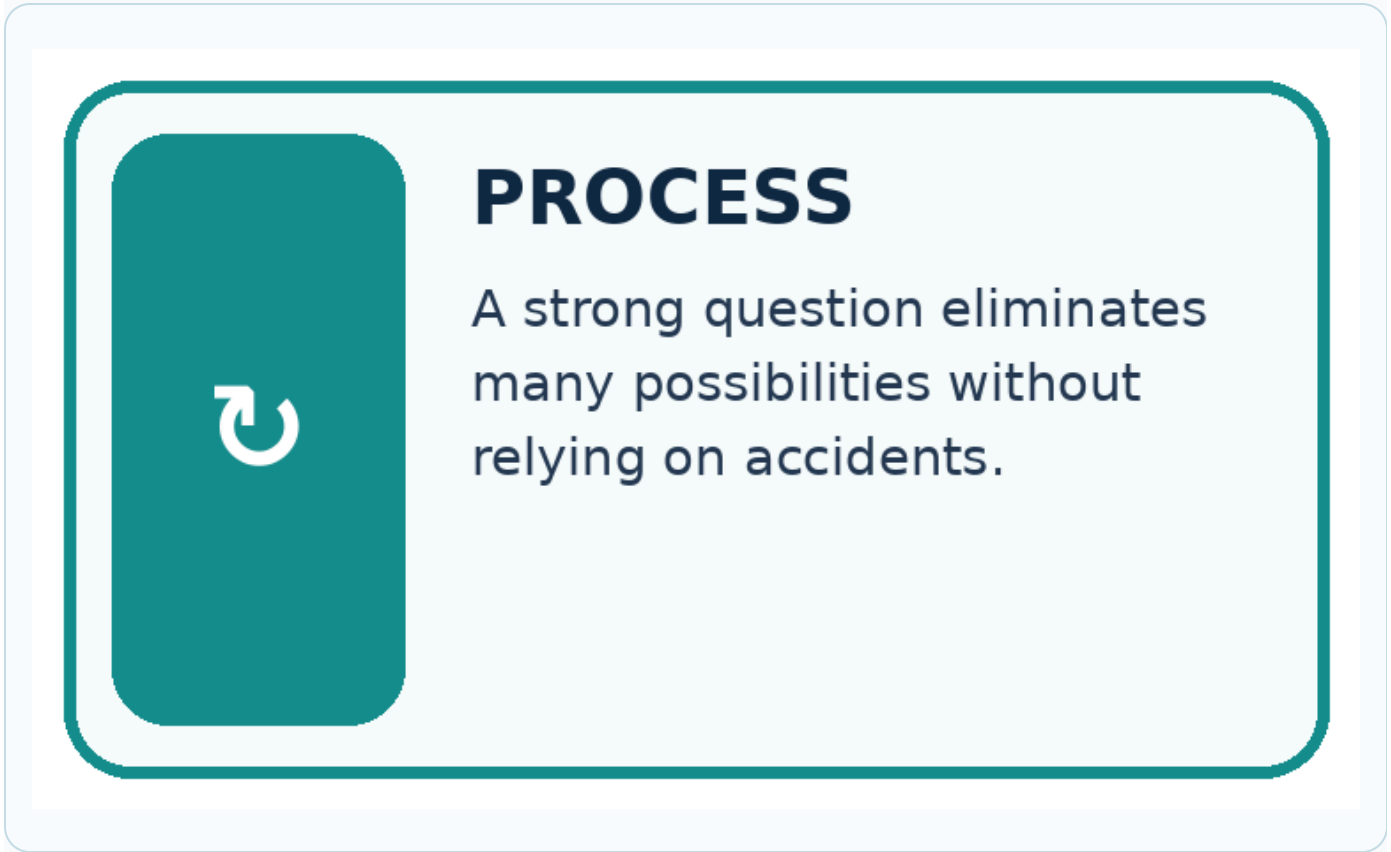
VISUAL **MECHANISM**

Input → **Process** → **Output**

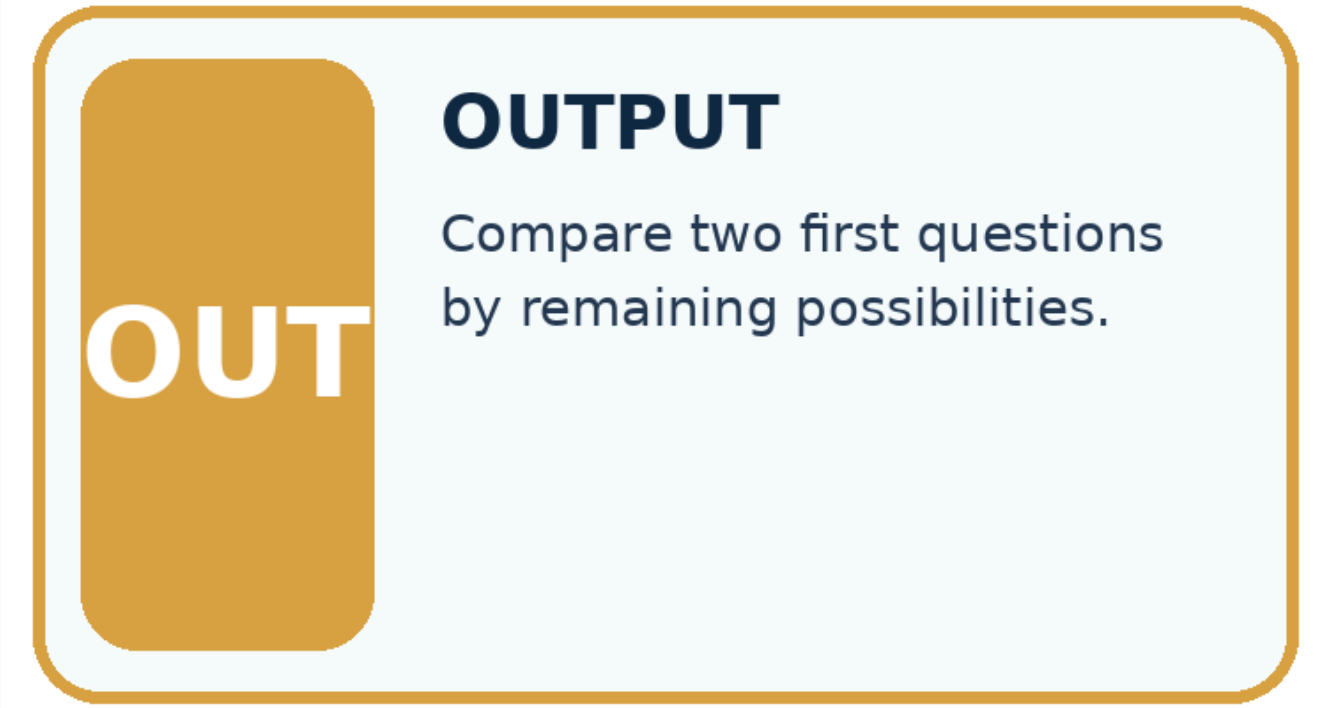
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compare two first questions by remaining possibilities.

↓


↻
PROCESS
A strong question eliminates many possibilities without relying on accidents.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

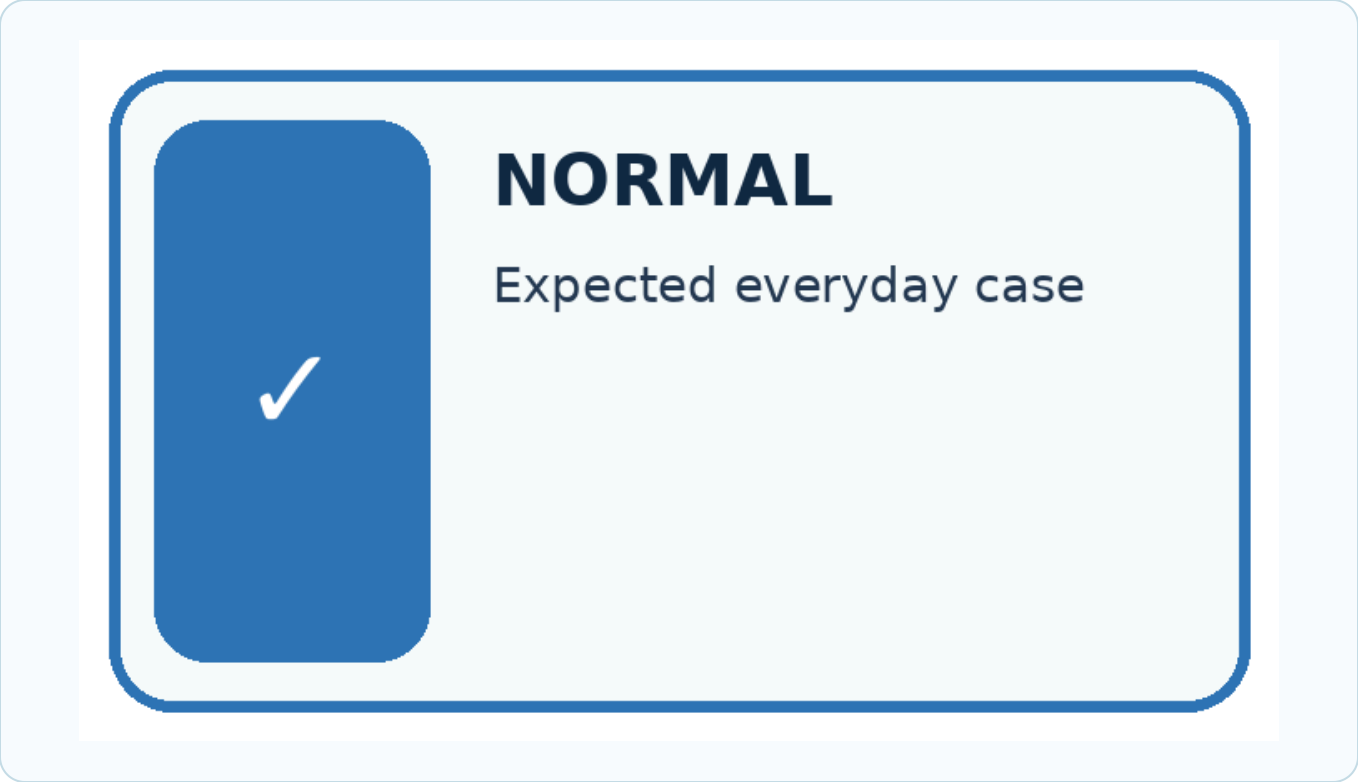
Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

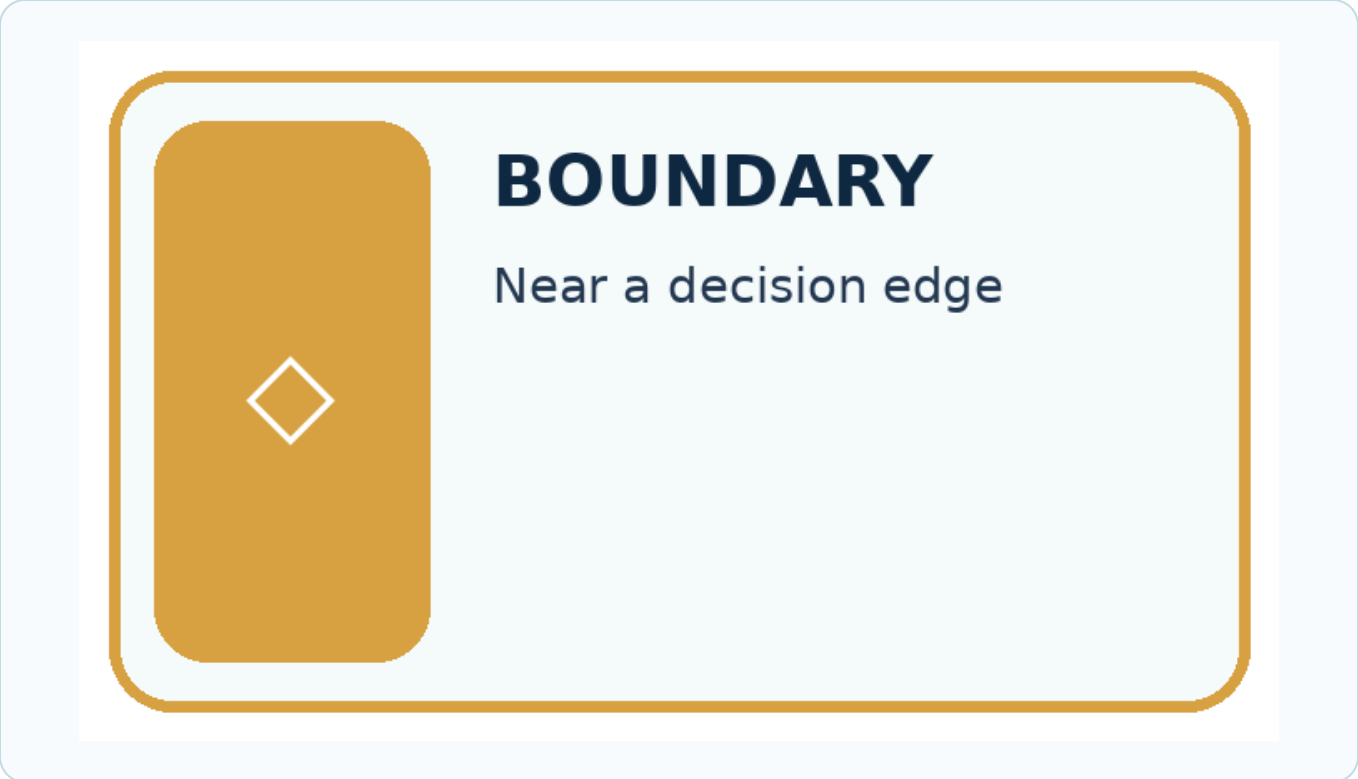
Choose a question for the current possibility set

> The value of a question can change after an answer.

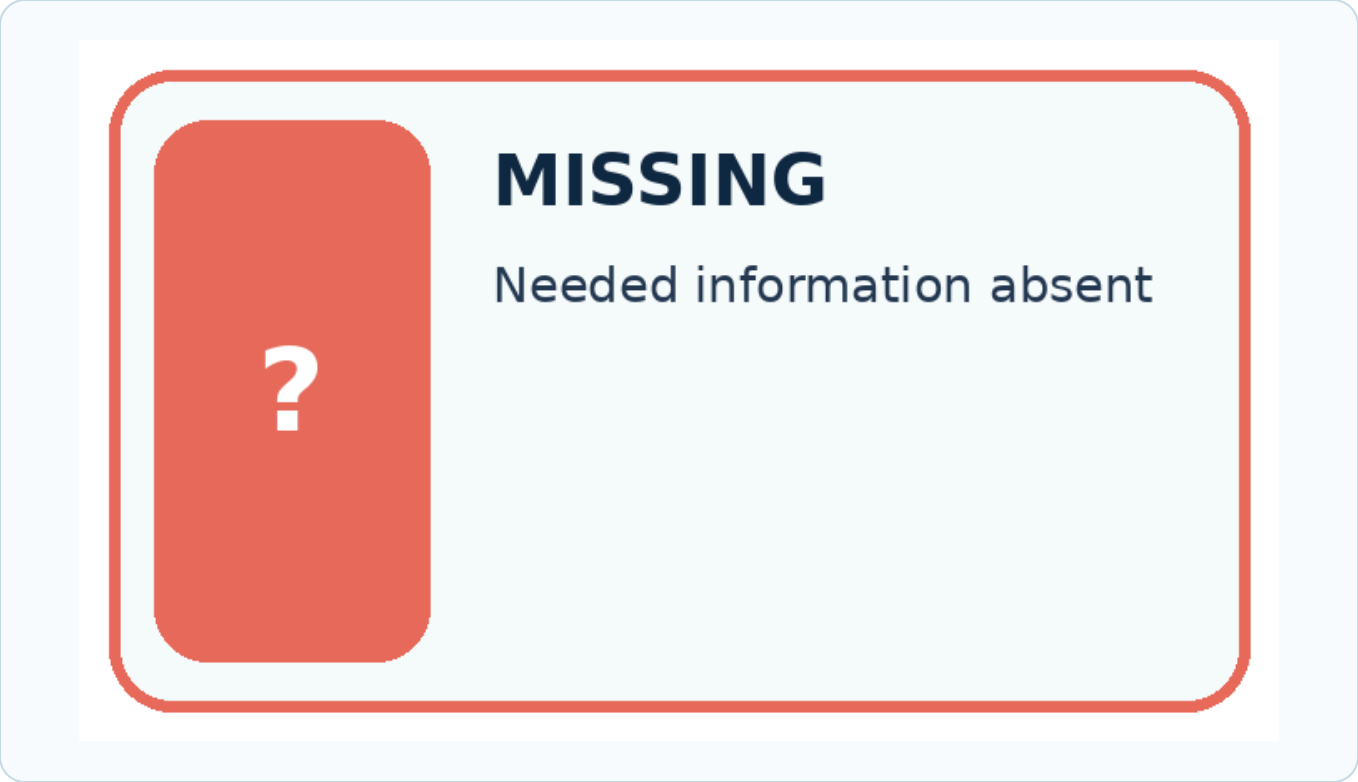
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Predict the supplied changed case

- Change: The current possibility set contains only the four mystery cards.

▼ Check after predicting

Prediction: Asking mystery now leaves all four; it no longer narrows the set.

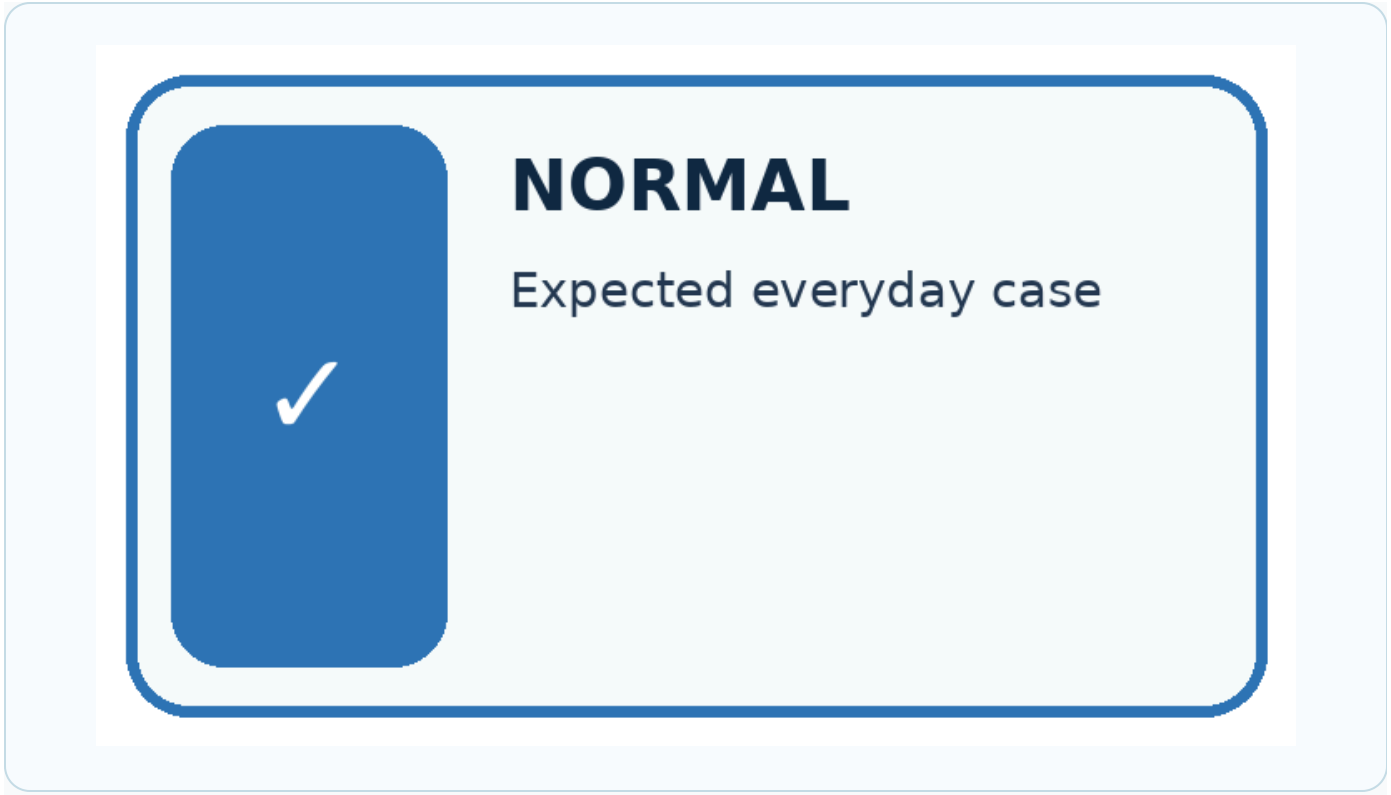
Reason: Question value depends on the remaining candidates, not just on its wording.

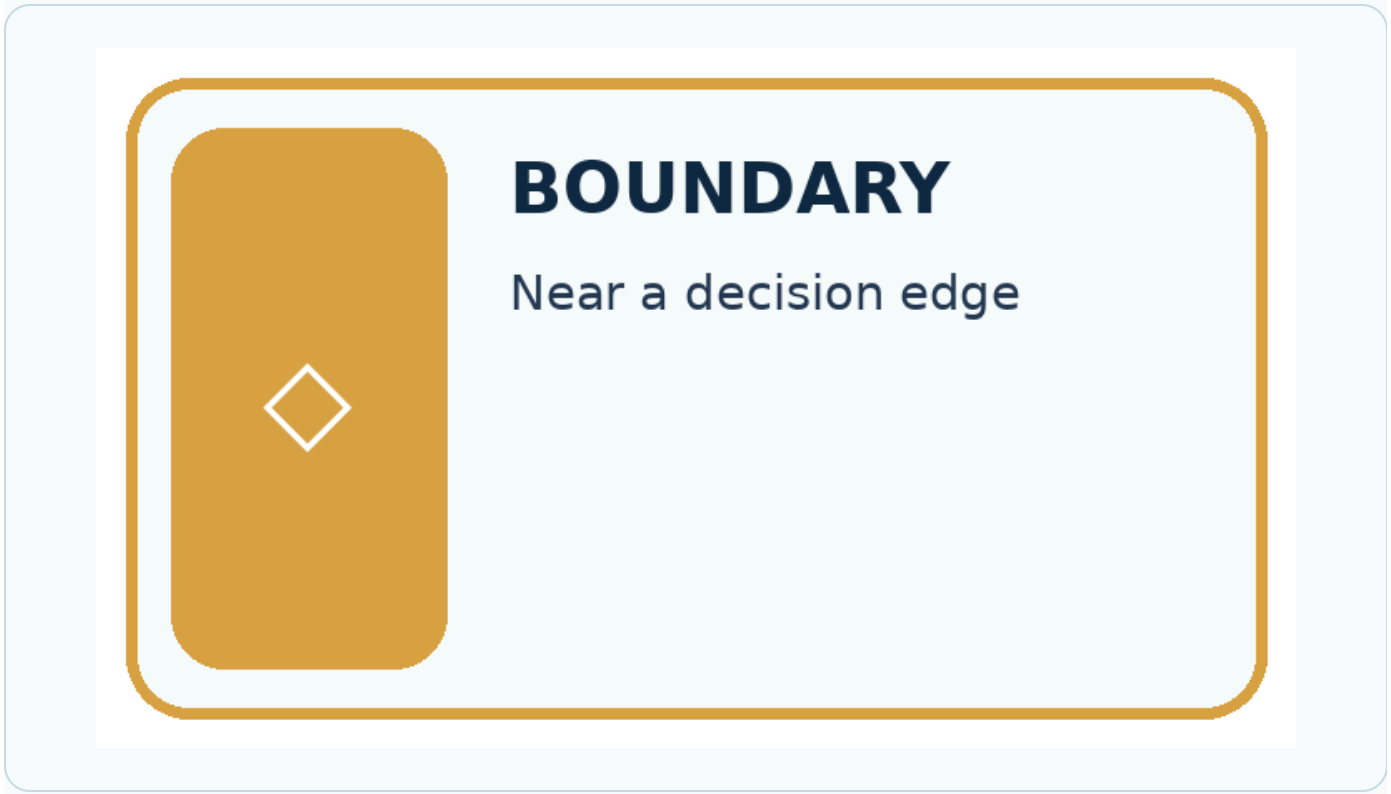
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Follow the eight-card worst-case comparison

> Keep this identification game distinct from last week’s fitted-tree review.

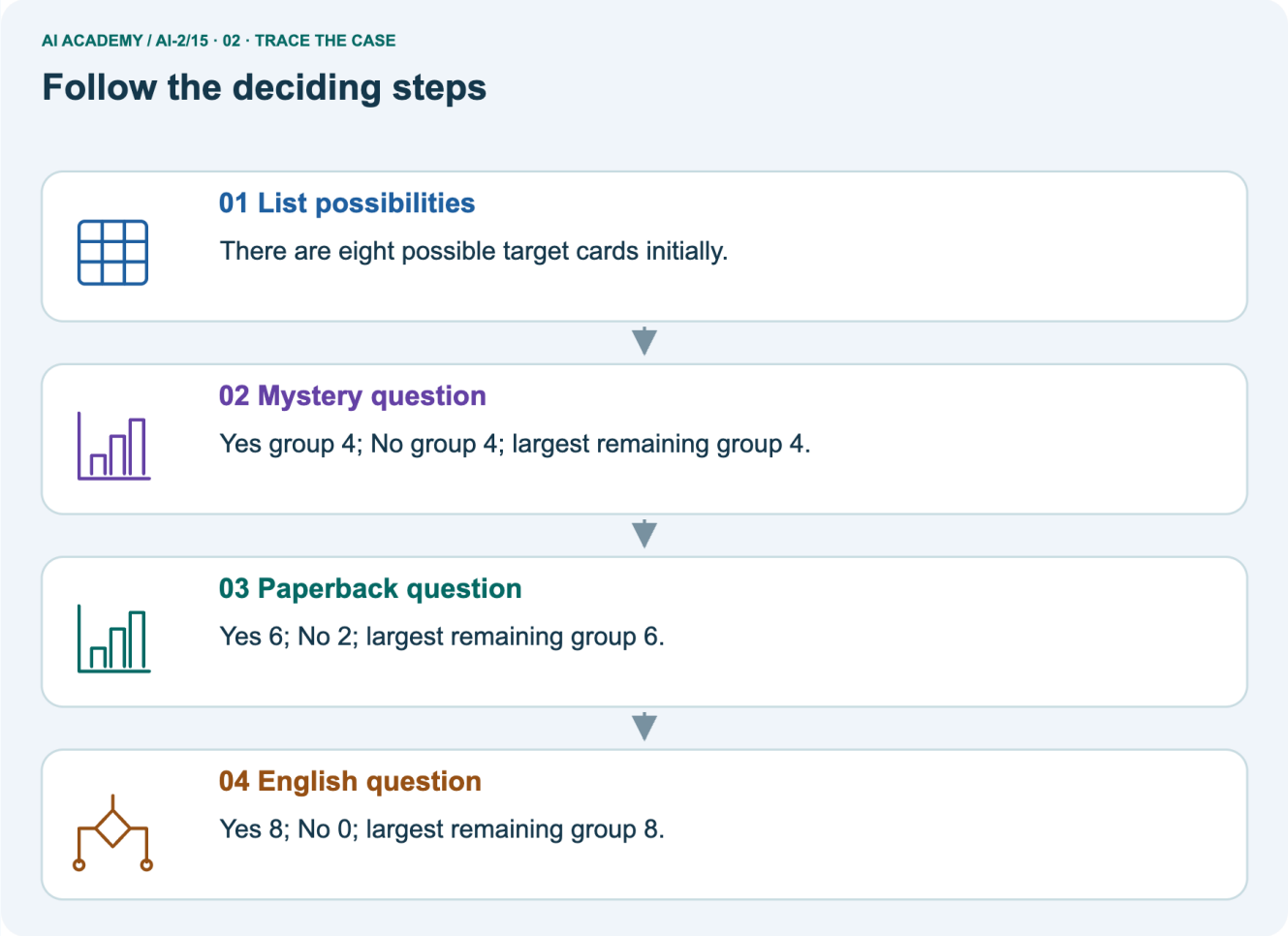
State the game and candidate questions

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

Trace all three branch counts



Follow the labelled picture with the matching explanation. Open the vector diagram

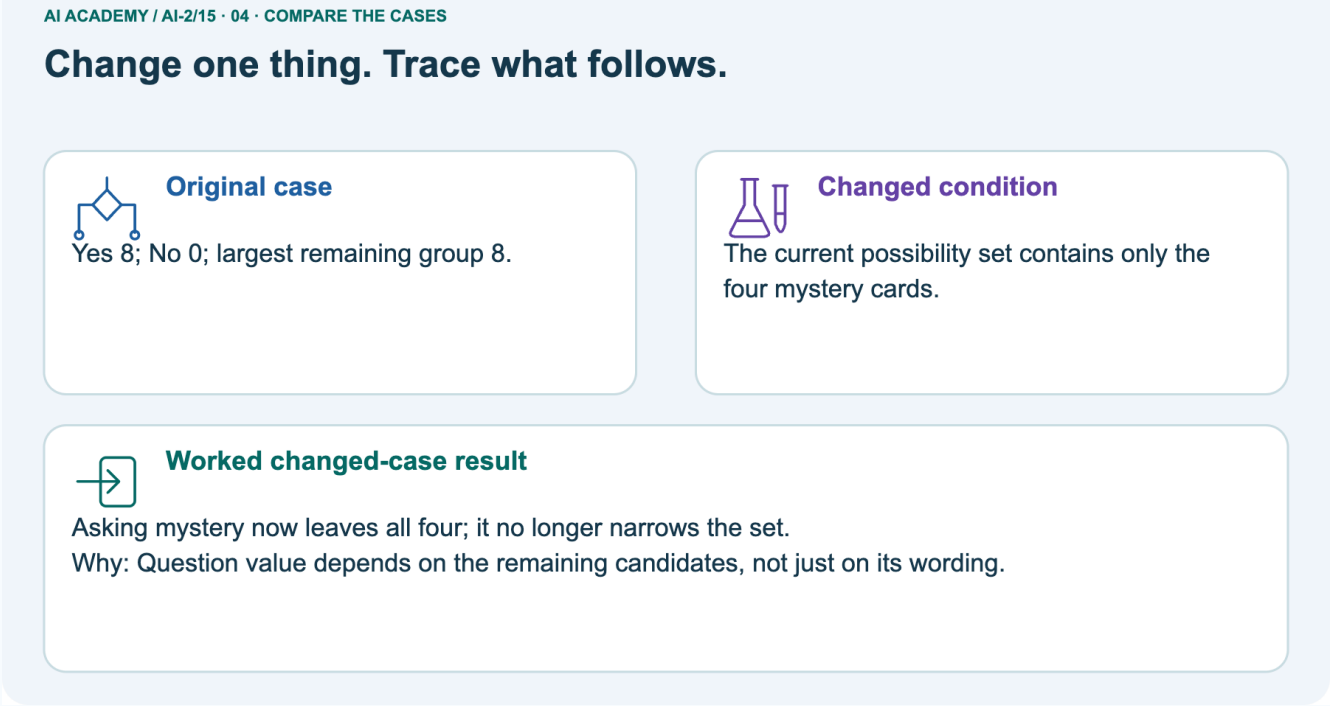
Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Predict the changed four-mystery set

- Change one thing: The current possibility set contains only the four mystery cards. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why balance matters under this criterion

- Read the labelled picture: Choose the best original question and explain why its more balanced **split** matters here.
- Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 15 • WORKED EVIDENCE

1. Eight cards remain.
2. ‘Has wings?’ splits 4/4; worst branch leaves 4.
3. ‘Is red?’ splits 1/7; worst branch leaves 7.
4. For this set, wings is the more informative first question.

1. Eight cards remain.

2. ‘Has wings?’ **split**s 4/4; worst **branch** leaves 4.

3. ‘Is red?’ **split**s 1/7; worst **branch** leaves 7.

4. For this set, wings is the more **informative** first question.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

> **Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.**

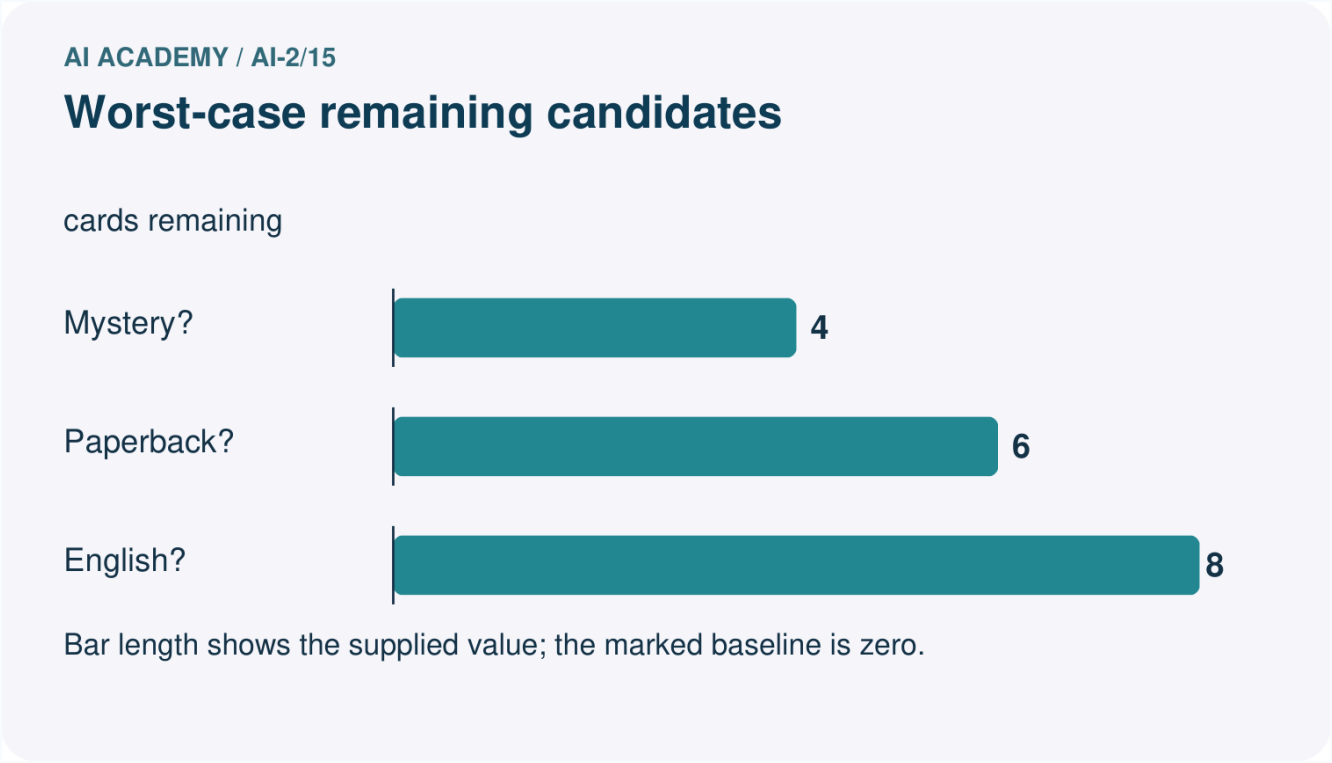
Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Choose Informative Questions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/15

Choose Informative Questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. List possibilities

There are eight possible target cards initially.



2. Mystery question

Yes group 4; No group 4; largest remaining group 4.



3. Paperback question

Yes 6; No 2; largest remaining group 6.



4. English question

Yes 8; No 0; largest remaining group 8.

CHANGE ONE THING

The current possibility set contains only the four mystery cards.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. List possibilities	There are eight possible target cards initially.
2. Mystery question	Yes group 4; No group 4; largest remaining group 4.
3. Paperback question	Yes 6; No 2; largest remaining group 6.
4. English question	Yes 8; No 0; largest remaining group 8.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Keep the criterion and its limits visible

> Balanced possibilities and low classification error are different goals.

What the branch counts establish

A 4/4 split leaves at most4; a6/2 split can leave6, and an8/0 split cannot narrow the original eight cards. This ranking answers a worst-case count question. It does not show that the answer is truthful, the tags are accurate, the question is equally cheap to ask, or the expected number of later questions is minimal. A different probability or cost model needs an explicitly different calculation.

▼ Week 14 review · Training evidence and validation limits

Record the two candidate questions, **branch** label counts, total errors, tie rules, **depth** limit, stored tree and validation **prediction**s.

Retain all rows, including the blocked example in the near=0 **branch**; removing a disagreement would change the **evidence**.

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset. This is retained Week 14 tree-fitting review. The next Week 16 lesson is Overfitting Intuition; do not use this excerpt as a nearest-neighbour preview.

> **A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The most unusual question is always the most informative. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Recount after an answer changes the possible targets

> Use the actual remaining cards, not a remembered headline.

Practise with the supplied table

On the original eight-card table, count Yes and No for paperback? before comparing your result. Now restrict to M1–M4 and recount each question. Do not carry the original6/2 binding split into a four-card population. English? remains constant, and mystery? has become constant too. A Yes to paperback? within M1–M4 still leaves three candidates, so it does not identify the target by itself.

Challenge a repeated constant question

- Repair this explanation before discussion: Mystery was the best first question, so it must remain the best question after we already know the target is a mystery. Point to the current cards and calculate the relevant two branch sizes.

Explain the original pictured comparison

Choose the best original question and explain why its more balanced **split** matters here.

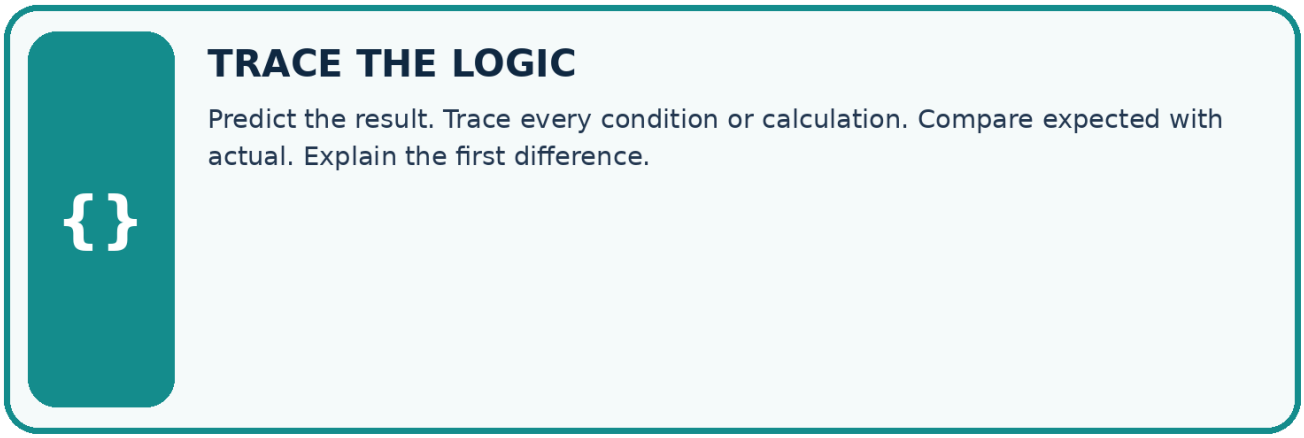
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Do not assume near=1 must always predict blocked. If a new **training** set gives that **branch** a clear majority, this same learner will store clear there. A feature name is not the answer.

Use the **training** labels belonging to that **branch**.

Adding more questions can fit finer **training** details but does not guarantee better new-case performance. Record any **depth** choice made during development, and keep final test data separate.

A **leaf** majority describes this **training** group; it is not a promise about every future example.

Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example "**Fitting** a decision tree". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Fitting** a decision tree. Proposed explanation: "A near **branch** must predict blocked". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit the original and changed possibility sets

> Keep predicted counts and observed paper/code output distinct.

Count, compare, restrict and recount

1. Use the complete eight-card table. Write predicted Yes, No and largest-branch counts for all three questions, and verify Yes+No=8 each time. Execute the supplied audit or label your work a paper count. Keep the chosen question and criterion. Then use only M1–M4, recount all three with Yes+No=4 and compare the chosen question again. Keep both populations and results visible. Ask a partner to check each ID belongs to exactly one branch. Record an inaccurate-tag or untruthful-answer limitation; do not fabricate a real user response. Reserve the new ten-card pet/space case for independent W4.

Keep the record and explain the changing choice

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

For each proposed question, count yes/no groups and the largest remaining group. Choose, justify, then test on a changed card set.

Use the complete eight-card catalogue table. Count both branches and their maximum for each question, select by the declared worst-case rule, then restrict to four mystery cards and recount. Preserve the current population and criteria beside each result.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare two questions on ten new cards

> Use the supplied counts and the largest-remaining-group criterion.

The independent pet/space case

Ten cards have six pet subjects and four space subjects; all are green. Compare asking “green?” with asking “pet?” to identify a card. Give the largest remaining group and one limitation.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Cards are two cats and two dogs, all red. Which question helps separate species: is it red or does it bark? State your assumption.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Ask first about preferences that remove the largest number of unsuitable books.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Six cards contain three animal and three space books; all have blue covers. Compare asking 'blue cover?' with 'animal subject?'. Is the better question guaranteed to work on inaccurately tagged books?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Choose recommender questions for a declared purpose

> Do not silently turn multiple acceptable books into one hidden target.

This week’s practical portfolio contribution

For an invented reader and catalogue, state the purpose of a question before ranking it. In a hidden-target game, minimise the largest possible remaining set. In a recommender, a reader may accept several subjects or not know a preference; truthful answers need not identify one hidden book. Declare allowed answers and review handling, avoid unnecessary personal questions, and record why the chosen question helps the actual task. Do not claim a worst-case split guarantees satisfaction.

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details.

Keep the evidence and human decision visible

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Ask first about preferences that remove the largest number of unsuitable books.

Evidence for your project

Finish: justify the question using its current branches

> State the current population, count criterion and one limitation.

Explain without notes

- Close your notes. Explain this goal in your own words: Compare questions by their largest remaining group

> **My deciding step, evidence and one remaining question**

Plan a changed-set return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
informative	Providing evidence that helps distinguish relevant possibilities or reduce uncertainty.	
split	Separate a dataset into portions with different development or evaluation roles.	
remaining possibilities	The options still consistent with the evidence after some have been ruled out.	
efficiency	The resources required to achieve a result, such as time, memory or number of steps.	

Terms for the connected explanation

Word	Meaning
Root	The first question asked by a tree

Word	Meaning
Branch	A path selected by the answer to a question
Leaf	An endpoint that stores a prediction
Depth	The largest number of question steps from root to leaf

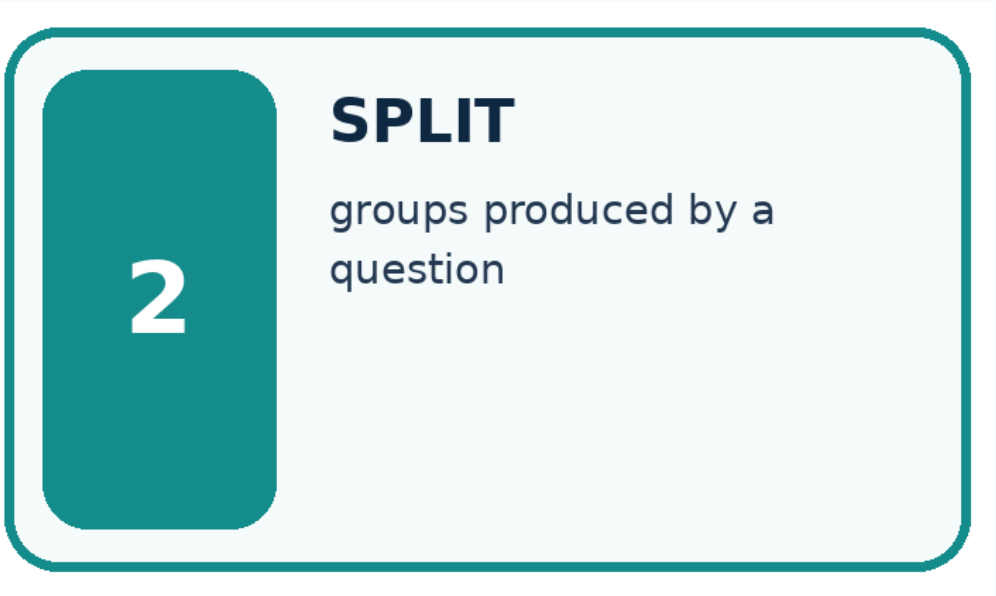
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
SPLIT
groups produced by a question


3
REMAINING POSSIBI
candidates still possible



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, **branch**, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

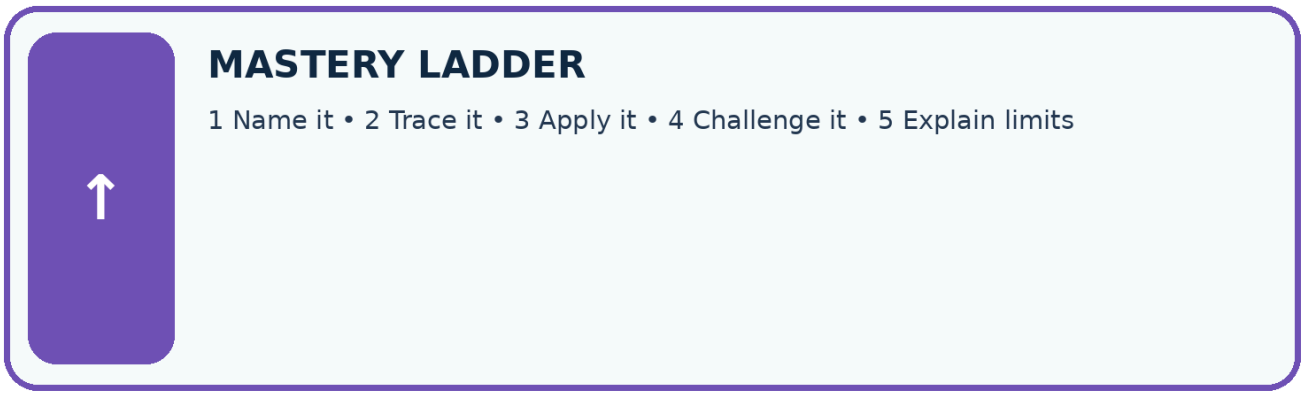
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Changed independent case is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

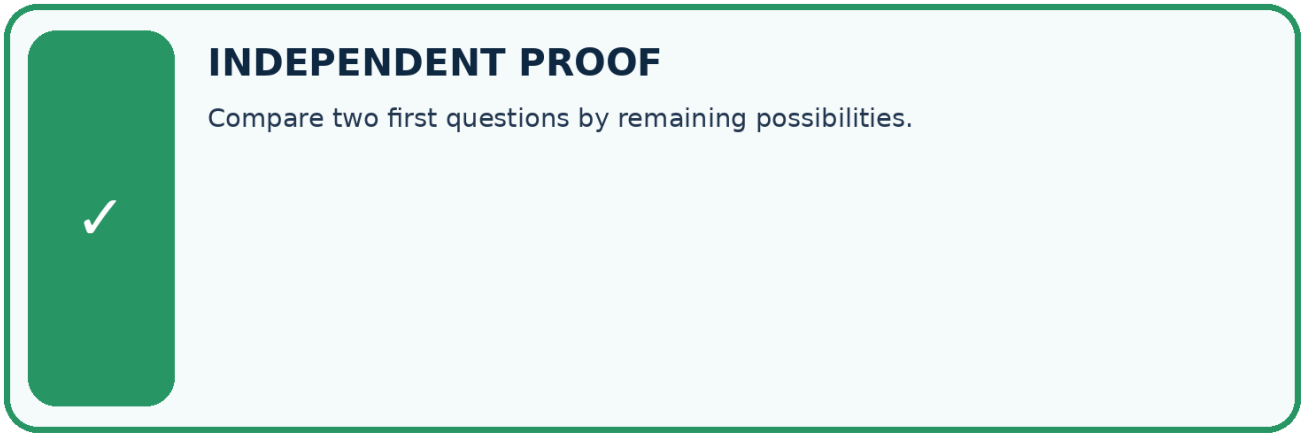
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Changed independent case is the selected independent case.



EXPLANATION CHECK Name the current possibility set, count both branches and compare the largest remaining groups under an explicit criterion.

Write or draw your reasoning and evidence.

▼ **Extra practice: Choose a question that separates the target**

REFERENCE · SECTION 13

Extra practice: Choose a question that separates the target

> Try the question before opening its worked explanation.

Read the supplied case and question

Eight blue cards contain four circles and four triangles; target is shape.

Which question is more useful: blue or circle?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

Evidence or stage	Value or result
Blue?	All eight stay together
Circle?	Four circles versus four triangles
Target	Shape, not colour

▼ Earlier wording, alternatives and picture transcripts

Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Choose a question that separates the target

Eight blue cards contain four circles and four triangles; target is shape.

Which question is more useful: blue or circle?

Circle separates target groups. Blue puts every card in one **branch**. Usefulness depends on the task and examples.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Compare questions by their largest remaining group	Explain the deciding step, show a trace or test, and state one limit.
Recount the split after the possibility set changes	Explain the deciding step, show a trace or test, and state one limit.
State assumptions and limits of a question-choice rule	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target.
- Which labelled step matches this detail? Yes 8; No 0; largest remaining group 8.
- 1. List possibilities
- 2. Mystery question
- 3. Paperback question

- 4. English question

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the first explanation

After your attempt, compare with the supported explanation: English question Yes 8; No 0; largest remaining group 8.

Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Changed-case question

- Changed-case practice: The current possibility set contains only the four mystery cards. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Asking mystery now leaves all four; it no longer narrows the set.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Asking mystery now leaves all four; it no longer narrows the set. Question value depends on the remaining candidates, not just on its wording.

Repair and save your work

Review Decision Trees (ai-2/14). Goal: Fit a **root** question and **leaf prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

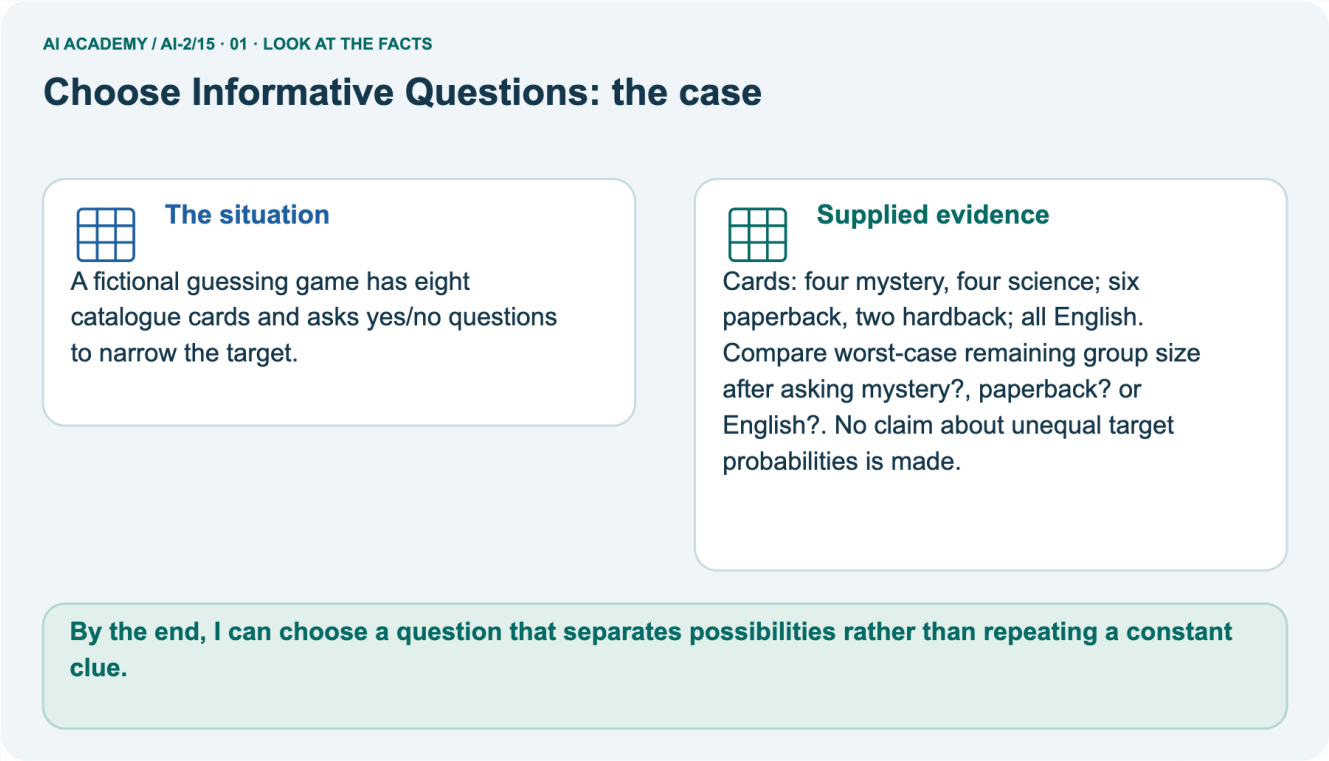
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Choose Informative Questions: the case



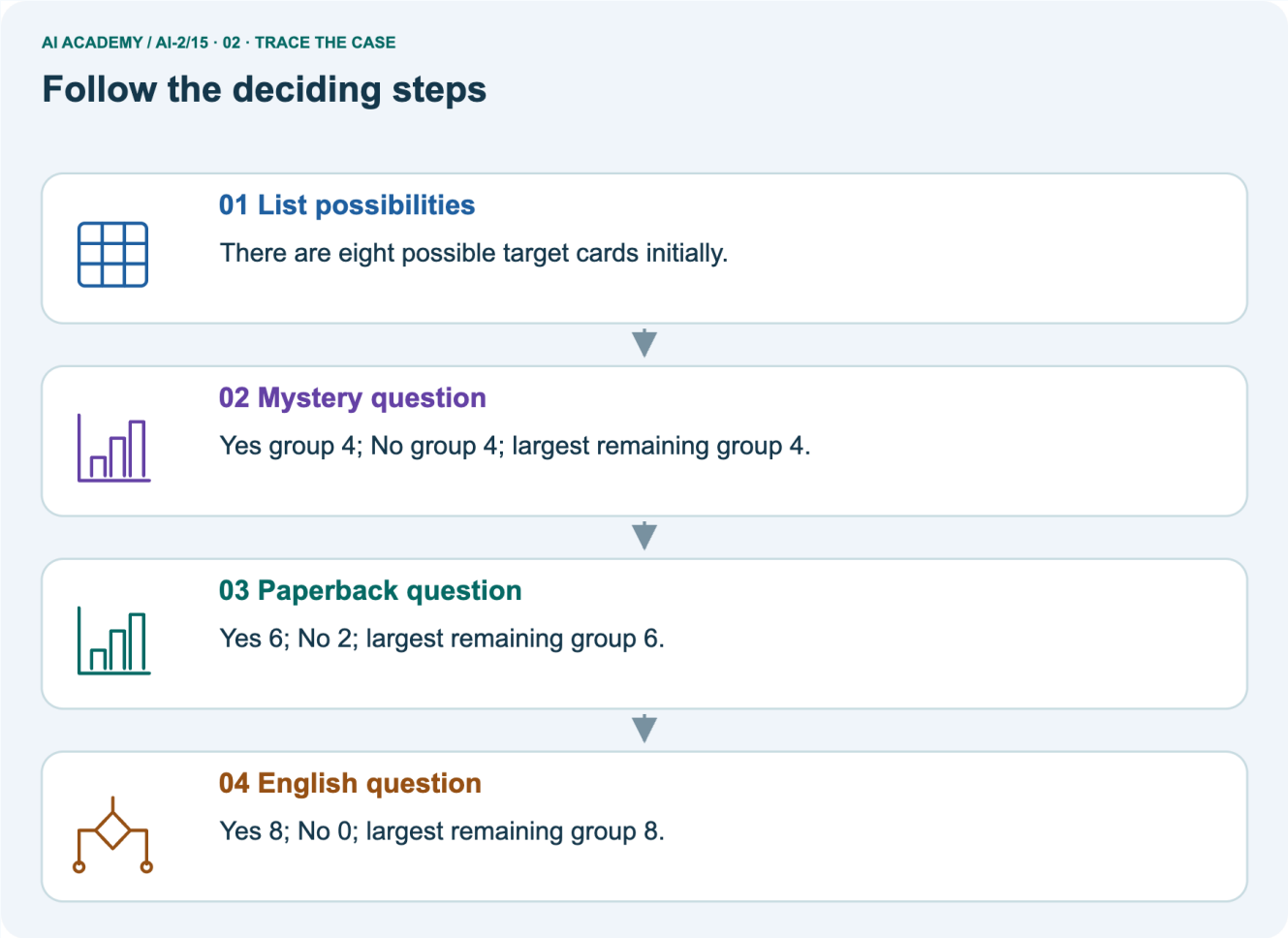
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional guessing game has eight catalogue cards and asks yes/no questions to narrow the target. Supplied facts: Cards: four mystery, four science; six paperback, two hardback; all English.

Compare worst-case remaining group size after asking mystery?, paperback? or English?. No claim about unequal target probabilities is made.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

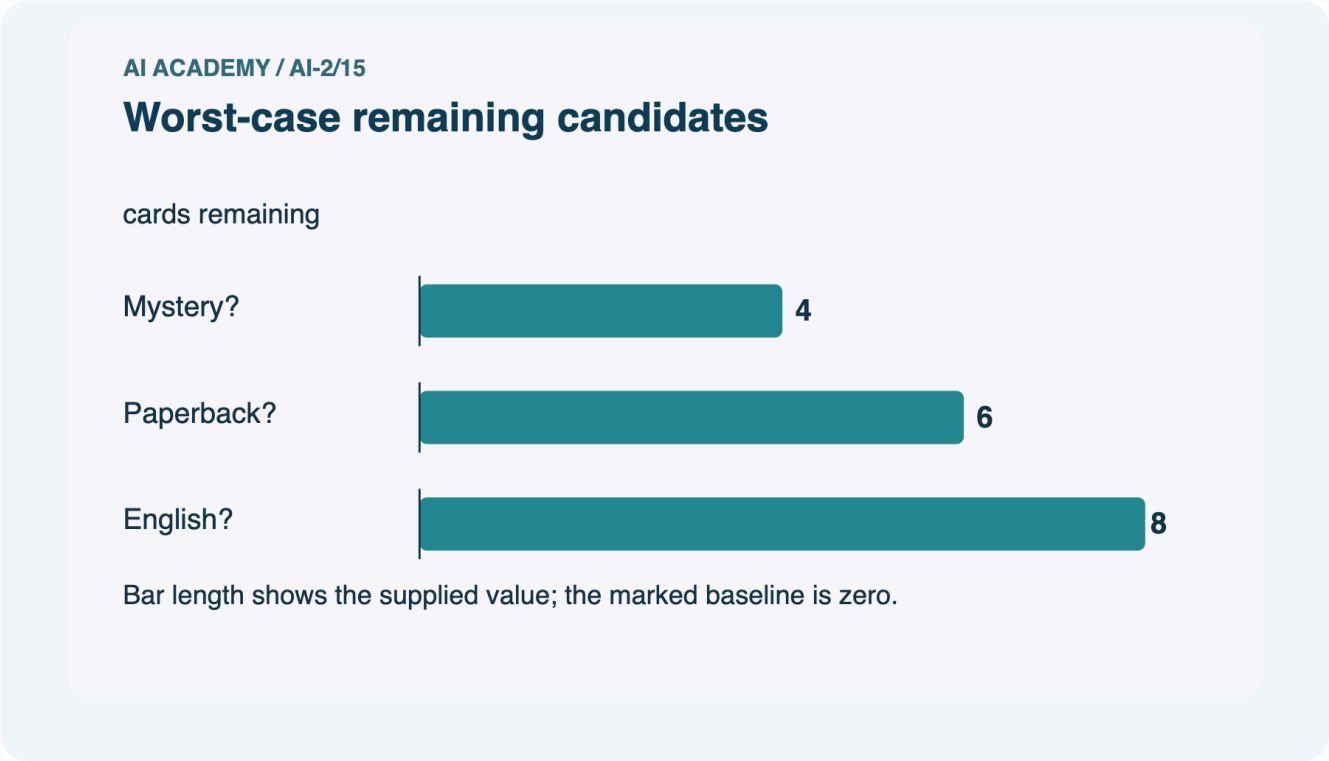


Follow the labelled picture with the matching explanation. Open the vector diagram

List possibilities: There are eight possible target cards initially. → Mystery question: Yes group 4; No group 4; largest remaining group 4. → Paperback question: Yes 6; No 2; largest remaining group 6. → English question: Yes 8; No 0; largest remaining group 8.

Point to each numbered stage and explain which recorded **input** it uses.

03. Worst-case remaining candidates

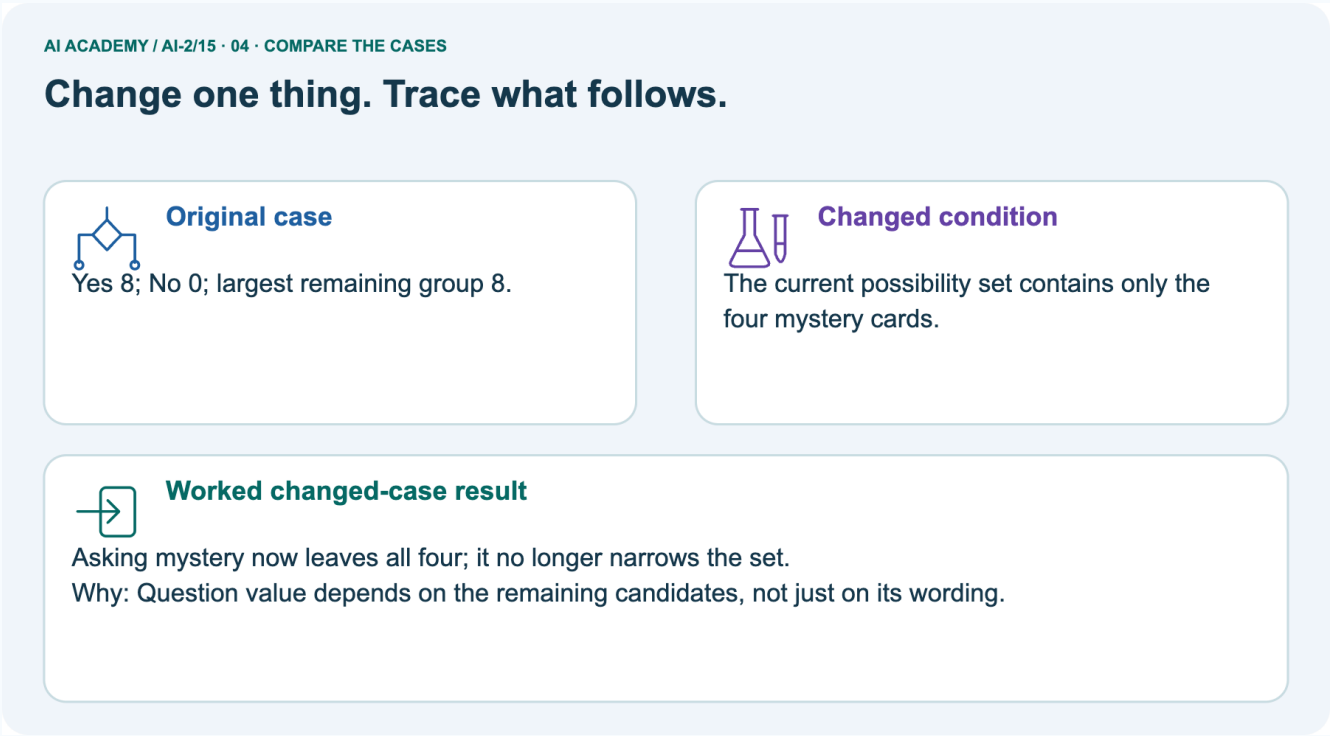


Follow the labelled picture with the matching explanation. Open the vector diagram

Worst-case remaining candidates. Units: cards remaining. Mystery?: 4. Paperback?: 6. English?: 8.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



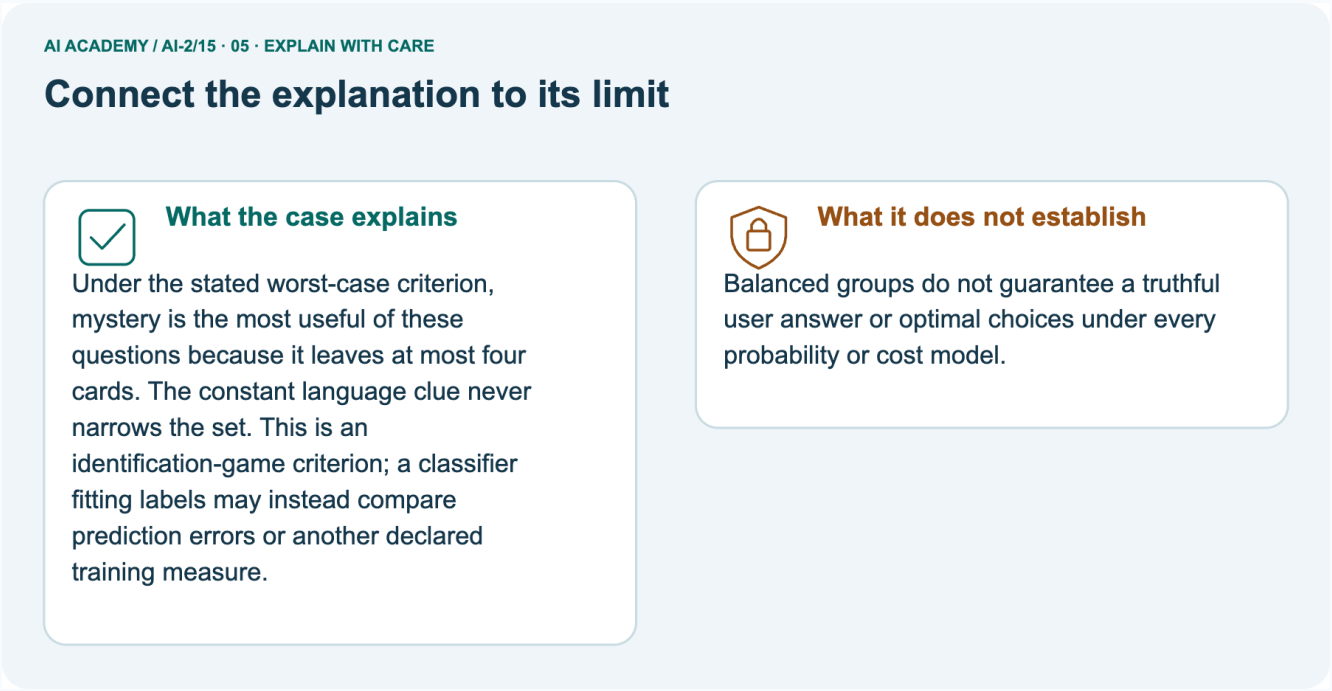
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Yes 8; No 0; largest remaining group 8. Changed condition: The current possibility set contains only the four mystery cards. Expected result: Asking mystery now leaves all four; it no longer narrows the set.

Reason: Question value depends on the remaining candidates, not just on its wording.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

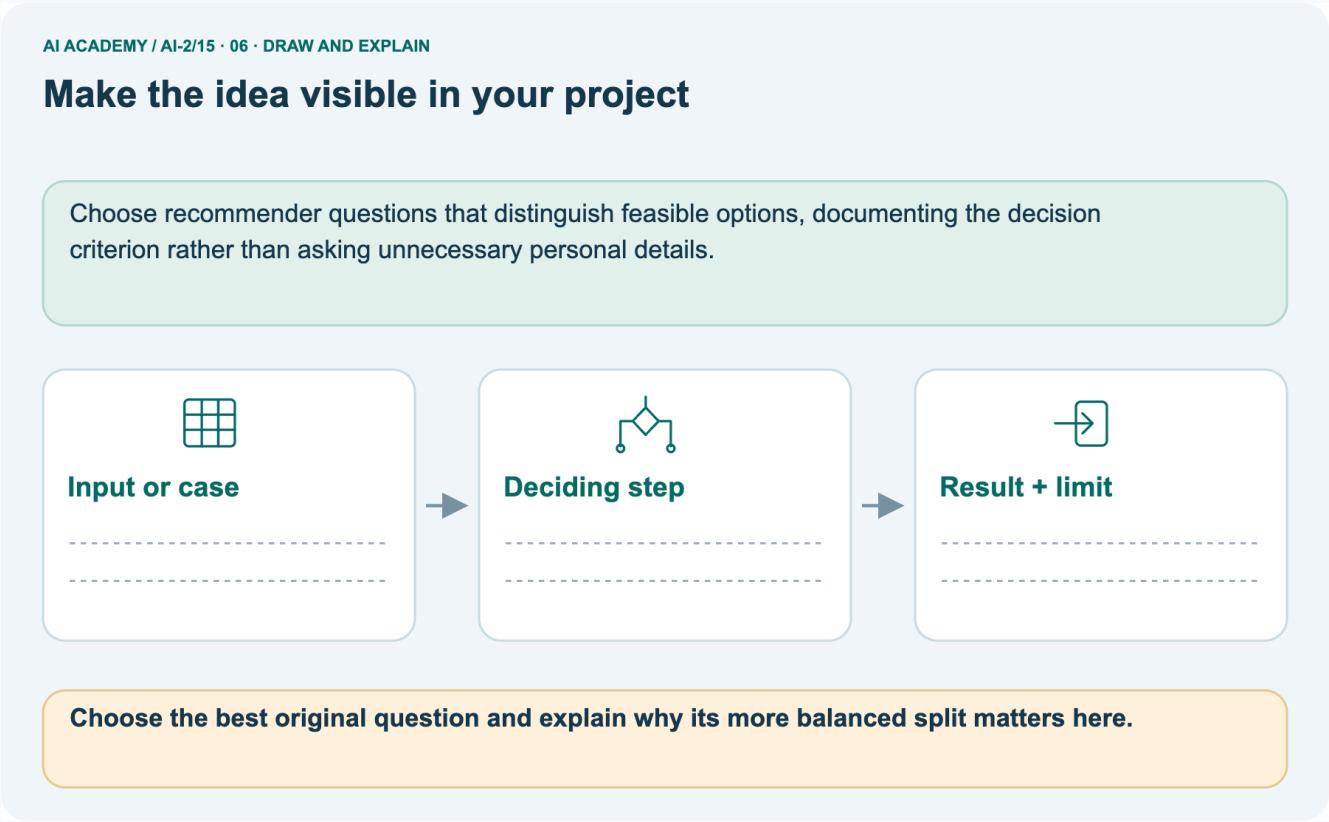
Under the stated worst-case criterion, mystery is the most useful of these questions because it leaves at most four cards. The constant language clue never narrows the set.

This is an identification-game criterion; a classifier **fitting** labels may instead compare **prediction** errors or another declared **training** measure.

Limit: Balanced groups do not guarantee a truthful user answer or optimal choices under every probability or cost model.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose recommender questions that distinguish feasible options, documenting the decision criterion rather than asking unnecessary personal details. Draw your **input**, deciding step, result and limit.

Supported question: Choose the best original question and explain why its more balanced **split** matters here.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Choose Informative Questions: the case

02. Follow the deciding steps

03. Worst-case remaining candidates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.